

ASTRATRACK — System Architecture

Engineering Specification, Flow Diagrams & Kinematics

ASTRATRACK: System Architecture & Design Specification

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System Architecture: Multi-Tier Deterministic Closed-Loop Simulation & Tracking Engine

1. Architectural Principles

ASTRATRACK is architected around five core principles essential for mission-critical aerospace simulation:

- 1. Strict Modularity & Decoupled Layers:** Physical simulation, perception, state estimation, control, and user presentation exist in isolated, interface-driven packages.
- 2. Bit-Exact Determinism:** Master-seed random number generation ensures that any simulation scenario can be recreated bit-for-bit, enabling scientific reproducibility.
- 3. Air-Gapped Self-Containment:** The simulation possesses zero dependencies on cloud APIs, internet connectivity, or proprietary runtime services.
- 4. Multi-Rate Operational Bandwidth:** The tracking control loop executes at up to 120 Hz, perception at 30–60 Hz, and telemetry visualization at 60 Hz.
- 5. Zero-Fake Data Guarantee:** All telemetry, metrics, and reports are directly derived from online mathematical evaluation of simulation states.

2. High-Level System Component Diagram

The diagram below illustrates the major subsystems and their inter-relationships:

```

graph TD
    subgraph SIMULATION_CORE [Simulation Core]
        W[3D World & Target Kinematics] --> C[Virtual Camera Body]
        C --> D[10-Channel Disturbance Engine]
    end

    subgraph PERCEPTION_TIER [Perception Tier]
        D -->|Degraded Frame| P[Perception Engine: AI / Classical]
    end

    subgraph ESTIMATION_AND_TRACKING [Estimation & Tracking Tier]
        P -->|Raw Centroid & Conf| FSM[Target-Loss & Re-Acquisition FSM]
        P -->|Observations| KF[6-State Kalman Filter]
        KF -->|State Est & Covariance| MP[Multi-Step Motion Predictor]
        FSM -->|Search Waypoint / Mode| KF
    end

    subgraph CONTROL_TIER [Control Tier]
        MP -->|Target Error| CC[Camera Controller: Dual-Axis PID]
        CC -->|Pan/Tilt Rate Cmds| C
    end

    subgraph METRICS_AND_UI [Telemetry & Presentation Tier]
        P --> MET[16-Metric Performance Collector]
        KF --> MET
        CC --> MET
        MET --> UI[Aerospace Engineering Dashboard]
    end

```

3. Data Flow & Processing Pipeline

The data flow pipeline executes sequentially on every simulation step Δt :

```

sequenceDiagram
    autonumber
    participant W as World / Target
    participant Cam as Virtual Camera
    participant Dist as Disturbance Engine
    participant Det as Perception (AI/CV)
    participant FSM as Tracking FSM
    participant Est as Kalman Estimator
    participant Ctrl as PID Controller
    participant Met as Metrics Collector

    W->>Cam: Update 3D Beacon Position (X, Y, Z)
    Cam->>Dist: Render Pristine Sensor Frame
    Dist->>Det: Inject Active Disturbance Channels (Noise/Turb/Blur)
    Det->>FSM: Extracted BBox, Centroid (u, v), Confidence
    FSM->>Est: State Transition (Tracking / Coasting / Spiral Search)
    Est->>Ctrl: Filtered State & Forward Predicted Coordinates
    Ctrl->>Cam: Dual-Axis Angular Rate Commands (dPan, dTilt)
    Cam->>Cam: Integrate Gimbal Kinematics & Update Actuators
    Cam->>Met: Log Tracking Error, FPS, and Lock Status

```

4. Finite State Machine (FSM) Transition Model

The target loss and re-acquisition engine operates as a 7-state finite state machine:

```
stateDiagram-v2
[*] --> SEARCHING
SEARCHING --> ACQUIRING: Target Detected (Conf >= Threshold)
ACQUIRING --> LOCKED: Confirmed for N consecutive frames
ACQUIRING --> SEARCHING: Lost before confirmation
LOCKED --> TRACKING: Optical lock achieved
TRACKING --> TARGET_LOST: Conf < Threshold for N_loss frames
TARGET_LOST --> PREDICTING: Kalman forward extrapolation active
PREDICTING --> REACQUIRING: Execute Archimedean spiral search
REACQUIRING --> LOCKED: Beacon redetected within search zone
REACQUIRING --> SEARCHING: Search timeout exceeded (Full reset)
```

5. Coordinate Systems & Spatial Transformations

ASTRATRACK implements four distinct right-handed coordinate frames:

5.1 World Frame $\mathcal{F}_w$

Origin at ground station optical terminal base. Coordinates $\mathbf{p}_w = [X_w, Y_w, Z_w]^T$. Defines absolute spatial positions of optical beacon and ground station platform.

5.2 Camera Body Frame $\mathcal{F}_c$

Origin at camera entrance pupil. Z_c aligned with optical boresight, X_c rightwards, Y_c upwards. Transformed from world frame via camera position $\mathbf{p}_{cam}$ and Euler rotation angles (ψ, θ) (Pan, Tilt):

$$\mathbf{p}_c = R_y(\theta) R_z(\psi) (\mathbf{p}_w - \mathbf{p}_{cam})$$

5.3 Pixel Sensor Plane $\mathcal{F}_p$

2D discrete sensor array $(W \times H)$ with origin at top-left pixel. Pinhole perspective projection with focal length f_x, f_y and principal point (c_x, c_y) :

$$u = f_x \frac{X_c}{Z_c} + c_x, \quad v = f_y \frac{Y_c}{Z_c} + c_y$$

5.4 Actuator Angular Rate Space $\mathcal{F}_a$

Translates pixel error into camera gimbal angular rate commands $(\dot{\psi}, \dot{\theta})$:

$$\dot{\psi}_{cmd} = \text{PID}_{pan}(u - c_x), \quad \dot{\theta}_{cmd} = -\text{PID}_{tilt}(v - c_y)$$

6. Mathematical Formulations

6.1 Discrete Kalman Filter (Constant Acceleration CA Model)

- **State Vector:** $\mathbf{x} = [x, y, v_x, v_y, a_x, a_y]^T$

- **Continuous White Noise Acceleration (CWNA) Covariance:**

$$\mathbf{Q} = \begin{bmatrix} \mathbf{Q}_{1D} & \mathbf{0} & \mathbf{0} & \mathbf{Q}_{1D} \end{bmatrix}, \quad \mathbf{Q}_{1D} = q_c \begin{bmatrix} \frac{\Delta t^5}{20} & \frac{\Delta t^4}{8} & \frac{\Delta t^3}{6} & \frac{\Delta t^2}{2} \\ \frac{\Delta t^4}{8} & \frac{\Delta t^3}{3} & \frac{\Delta t^2}{2} & \Delta t \end{bmatrix}$$

- **Measurement Update with Innovation:**

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{H} \hat{\mathbf{x}}_{k|k-1}$$

$$\mathbf{S}_k = \mathbf{H} \mathbf{P}_{k|k-1} \mathbf{H}^T + \mathbf{R}$$

$$K_k = P_{k|k-1} H^T S_{k-1}^{-1}$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k-1} + K_k \mathbf{y}_k$$

$$P_{k|k} = (I - K_k H) P_{k|k-1} (I - K_k H)^T + K_k R K_k^T \quad \text{(Joseph Stabilized Form)}$$

6.2 Dual-Axis PID Control with Anti-Windup & Derivative Filter

• Integral Accumulation:

$$I(k) = I(k-1) + e(k) \Delta t$$

$$I(k) = \text{clamp}\left(I(k), -\frac{u_{\max}}{K_i}, \frac{u_{\max}}{K_i}\right)$$

• Filtered Derivative:

$$D(k) = \alpha \frac{e(k) - e(k-1)}{\Delta t} + (1 - \alpha) D(k-1)$$

• Total Output with Dead-Zone:

$$u(k) = \begin{cases} 0 & \text{if } |e(k)| \leq e_{\text{dead}} \\ \text{clamp}(K_p e(k) + K_i I(k) + K_d D(k), -u_{\max}, u_{\max}) & \text{otherwise} \end{cases}$$

6.3 Archimedean Spiral Re-Acquisition Path

When entering **REACQUIRING**, search coordinates are generated parameterized by angular displacement $\theta(t) = \omega t$:

$$r(t) = r_0 + \frac{v_{\text{search}}}{2\pi} \theta(t)$$

$$x_{\text{cmd}}(t) = x_{\text{last}} + r(t) \cos(\theta(t)), \quad y_{\text{cmd}}(t) = y_{\text{last}} + r(t) \sin(\theta(t))$$

Guaranteeing uniform spatial scanning of the uncertainty ellipse without blind spots.

7. Threading, Timing, and Determinism

- **Synchronous Simulation Stepping:** All submodules advance via a unified discrete time-step $\Delta t = 1/\text{FPS}$.
- **Master PRNG Architecture:** High-quality Mersenne Twister / PCG streams keyed by master seed, ensuring zero cross-run divergence.
- **UI Decoupling:** The UI rendering loop is decoupled from simulation updates using double-buffered telemetry state snapshots, eliminating UI thread jitter on simulation dynamics.